

STRENGTH OF MATERIAL

Self
confidence

Achieve
Daily Target

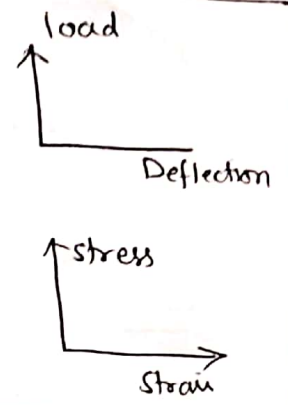
Multiple
revision

To The Point-ByDhyanPal(ESE'17AIR-179,GATE'18AIR-93,GATE'16AIR-145)



MEDICAL CAMERA

Toughness	area of Load-Deflection diagram till fracture
Resilience	_____ till elastic limit
modulus of toughness	area of stress-strain diagram till fracture
modulus of Resilience	_____ till elastic limit

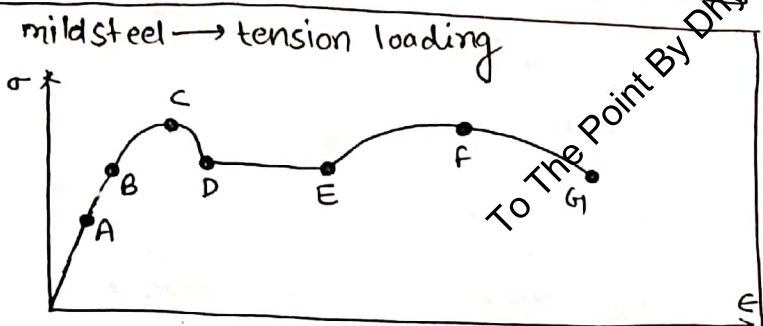


Homologous temp $\Rightarrow$ at which creep uncontrollable

Brinell hardness no = $\frac{P}{\frac{\pi D}{2} [D - \sqrt{D^2 - d^2}]}$

D $\rightarrow$ steel ball dia. }
 d $\rightarrow$ Indent dia } mm

Proof stress :- Stress required to cause a permanent extension equal to defined % of gauge length like 0.2 % of strain $\rightarrow$ whatever stress is known as proof stress.

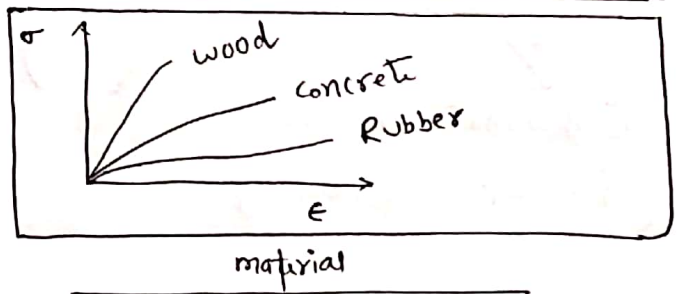


- A $\rightarrow$ limit of Proportionality (Hook's law) valid
- B $\rightarrow$ elastic limit
- C $\rightarrow$ upper yield point $\rightarrow$ not depends on length
- Depends on:
 - cross-sectional area
 - shape
 - equipment
- D $\rightarrow$ lower yield point
- E $\rightarrow$ strain hardening starts
- F $\rightarrow$ ultimate point
- G $\rightarrow$ fracture point

Presence of freigraphite $\rightarrow$ Material is Cast Iron

Note:- steel has no graphite

Percentage carbon $\uparrow \Rightarrow$ yield stress $\uparrow$
 But fracture strain $\downarrow$



material	
ductile	Brittle
(Post elastic strain $> 5\%$) Ex. lead mild steel Copper	($< 5\%$) Ex. Rubber glass cast Iron
note:- In Brittle material $\rightarrow$ NO plastic zone	

V.V.V. Imp

material

ductile

weak in shear

Brittle

weak in brittle

order :- shear < tensile ≤ compressive

(ductile material weakest in shear)

order :-

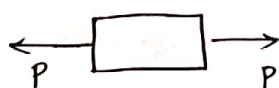
Tensile < shear < compressive

(Brittle material like concrete weak in tension)

Ductile material

Brittle material

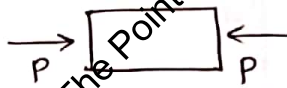
tensile loading



∴ ductile weak in shear ∴ fails in shear [shear max. at 45°]

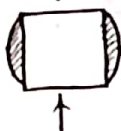
"CUP-CONE FRACTURE" failure plane @ 45°

Compressive loading



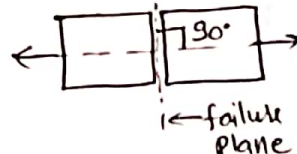
To The Point By AnyenPal

BULGING failure



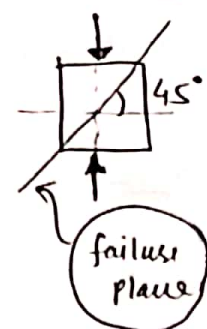
Tensile loading

∴ Brittle weak in tension hence 90° from longitudinal axis failure plane.



Compressive loading

∴ strong in compression hence fail in shear @ 45°



Torsion applied { basically Torque T }

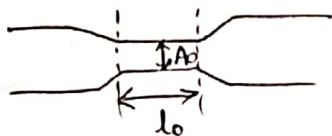
Ductile material shaft

Plane fracture

Brittle material shaft

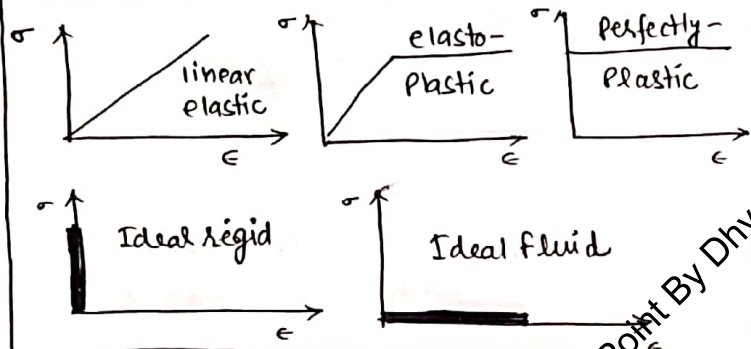
Helicoidal surface

Ductility measurement
gauge length = $5.65 \sqrt{A_0}$



Tenacity $\rightarrow$ Resist fracture under tensile load

Some stress-strain diagram :-



Viscoelastic material $\rightarrow$ time dependent stress-strain curve

Isotropic material $\rightarrow$ Identical properties in all direction.

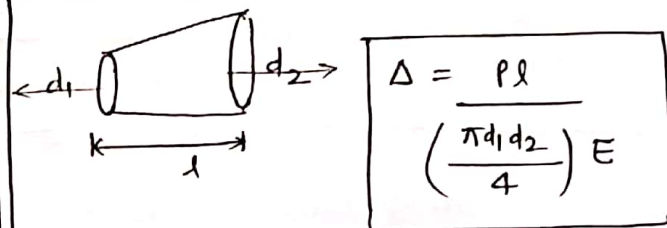
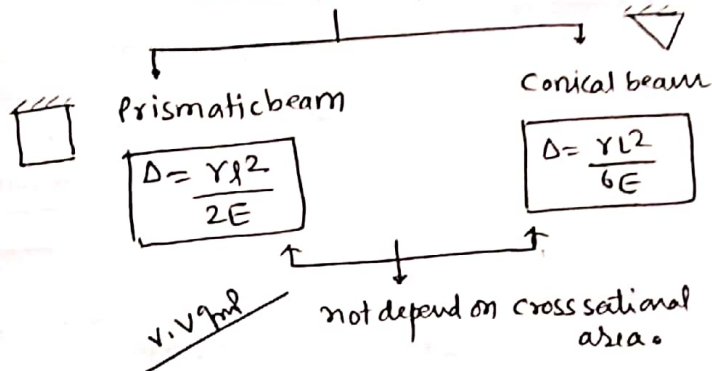
Homogeneous material $\rightarrow$ similar properties throughout volume.

material	no. of Independent elastic constant
homogeneous isotropic	2
orthotropic (wood)	9
Anisotropic	21

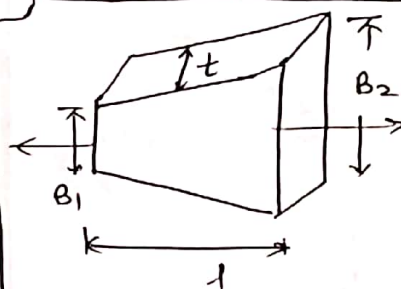
noti:- Anisotropic material $\rightarrow$ their properties changes when measured from different directions.

orthotropic $\rightarrow$ subset of Anisotropic material
 $\therefore$ their properties differ along 3 mutually direction.

Selfweight Deflation

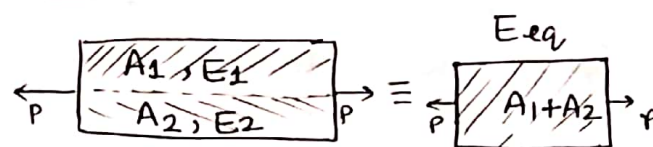


noti:- $\Delta_{actual} > \Delta_{avg. dia}$
 $d = \frac{d_1 + d_2}{2}$
 $+11.11\% \text{ error}$



$$\Delta = \frac{Pl \ln\left(\frac{B_2}{B_1}\right)}{(B_2 - B_1)tE}$$

Composite Bar :-



$$P = P_1 + P_2 \quad \text{--- (i)}$$

$$\Delta_1 = \Delta_2 \quad \text{--- (ii)}$$

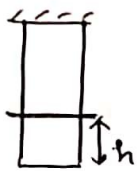
$$E_{eq} = \frac{A_1 E_1 + A_2 E_2}{A_1 + A_2}$$

$$\Delta_1 = \frac{P_1 L}{A_1 E_1}$$

$$\Delta_2 = \frac{P_2 L}{A_2 E_2}$$

$$P_1 = P \frac{A_1 E_1}{A_1 E_1 + A_2 E_2}$$

Impact loading :-



$$\sigma_{\text{dynamic}} = \sigma_{\text{static}} \left[1 + \sqrt{1 + \frac{2h}{\Delta_{\text{static}}}} \right]$$

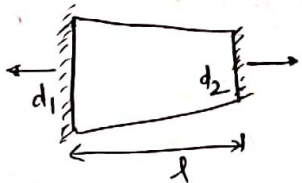
$$\text{Impact factor (IF)} = \frac{\sigma_{\text{dynamic}}}{\sigma_{\text{static}}} \geq 2$$

Sudden Impact $h=0 \therefore \text{Impact factor} = 2$

Dilation = sum of strain in all direction

$$\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

tapered bar :-



Temperature Increase

$$\Delta l \Delta T - \frac{Pl}{\left(\frac{\pi d_1 d_2}{4} \right) E} = 0$$

$$\therefore \sigma_{\text{max}} = \alpha T E \left(\frac{d_1}{d_2} \right)$$

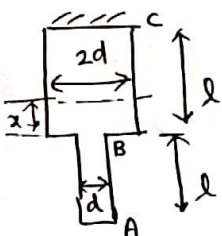
$$\sigma_{\text{min}} = \alpha T E \left(\frac{d_2}{d_1} \right)$$

Special case



$$\sigma = \alpha T E$$

Self wt conceptual :-



$$\Delta = \Delta_{AB} + \Delta_{BC}$$

$$\Delta_{AB} = \frac{Y_{AB} l_{AB}^2}{2 E_{AB}}$$

$\Delta_{BC} = \text{Self wt of BC} + \text{due to wt of AB}$

$$\frac{Y_{BC} l_{BC}^2}{2 E_{BC}} + \left[\frac{Y_{AB} \pi d_{AB}^2 l_{AB}}{4} \right] \times l_{BC}$$

wt. of AB

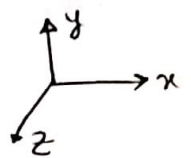
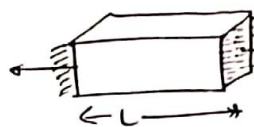
$$A_{BC} \cdot E_{BC}$$

$$\downarrow \frac{\pi}{4} (2d)^2$$

Method-1

Restraint problems

(i) Restrain from 1 side



$\sigma_x \rightarrow \text{generate}$

$\sigma_y \rightarrow \text{zero}$

$\sigma_z \rightarrow \text{zero}$

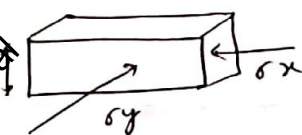
} $\therefore$ free

$\therefore \Delta$ in x direction zero { $\therefore$ fixed }

$$L \alpha \Delta T - \frac{\sigma_x}{E} l = 0$$

$$\sigma_x = \alpha T E$$

(ii) Restrain from 2 sides



$$\therefore \Delta x = 0$$

$$\Delta y = 0$$

$$\Delta x = l \alpha \Delta T - \frac{\sigma_x}{E} l + \frac{4 \sigma_y}{E} \times l = 0$$

$$\Delta y = b \alpha \Delta T - \frac{\sigma_y}{E} \times b + \frac{4 \sigma_x}{E} b = 0$$

$$\text{Solve } \sigma_x = \sigma_y = \frac{\alpha T E}{1 - 4}$$

(iii) Restrain from all 3 sides

$$\sigma_x = \sigma_y = \sigma_z = \frac{\alpha T E}{1 - 24}$$

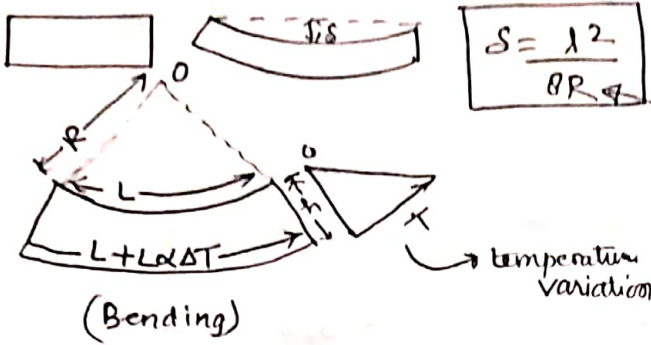
Δ_{BC}

Method-2

$$\int_0^l \left[Y_{AB} l_{AB} A_{AB} + Y_{A_x} \cdot x \right] dx$$

$$\frac{\pi}{4} (2d)^2 \cdot E_{BC}$$

A_{BC}



$$\delta = \frac{\lambda^2}{8R}$$

(Bending)

$$\frac{L}{R} = \frac{L + L\alpha\Delta T}{R + h} \quad \therefore R = h / \alpha T$$

$$\therefore \delta = \frac{\alpha T \lambda^2}{8h}$$

$$\mu = 0.25 \text{ to } 0.42 \text{ Elastic material}$$

$$\mu = 0.15 \text{ concrete}$$

$$E = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} \quad \text{modulus of elasticity}$$

$$G = \frac{\text{shear stress}}{\text{shear strain}} \quad \text{modulus of rigidity}$$

$$K = \frac{\text{direct normal stress}}{\text{volumetric strain}} \quad \text{Bulk modulus}$$

$$\mu = \frac{\text{lateral strain}}{\text{longitudinal strain}} \quad \text{Poisson ratio}$$

$$\mu = \frac{3K - 2G}{6K + 2G}$$

$$\frac{E}{3} < G < \frac{E}{2}$$

$$\text{Stress matrix} \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$

$$\text{Symmetric matrix} \\ \therefore \tau_{xy} = \tau_{yx} \\ \tau_{xz} = \tau_{zx} \\ \tau_{yz} = \tau_{zy}$$

$$\text{Strain matrix} \begin{bmatrix} \epsilon_{xx} & \frac{\phi_{xy}}{2} & \frac{\phi_{xz}}{2} \\ \frac{\phi_{yx}}{2} & \epsilon_{yy} & \frac{\phi_{yz}}{2} \\ \frac{\phi_{zx}}{2} & \frac{\phi_{zy}}{2} & \epsilon_{zz} \end{bmatrix}$$

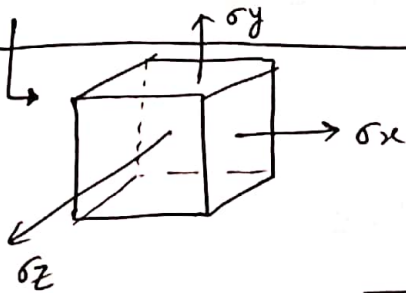
$$\epsilon_{xx} = \frac{\partial u}{\partial x} = \text{normal strain in } x \text{ direction} \\ \phi_{xy} = \text{shear strain} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

$$E = 2G(1 + \mu) = 3K(1 - 2\mu) = \frac{9KG}{3K + G}$$

$$\text{Volumetric strain } \frac{\Delta V}{V} = \epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z = \frac{\sigma_x + \sigma_y + \sigma_z}{E} (1 - 2\mu)$$



for perfectly plastic material $\Delta V = 0 \therefore \epsilon_v = 0 \therefore \mu = 0.5$

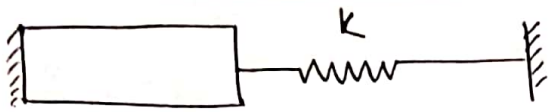


$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\mu \sigma_y}{E} - \frac{\mu \sigma_z}{E}$$

Same ϵ_y, ϵ_z

V.V.S.M

Wrought Iron $\rightarrow$ Purest form



$$\sigma = \frac{\alpha T E}{1 + \left(\frac{AE/L}{k} \right)}$$

$\frac{AE}{L} \rightarrow$ Axial stiffness

Rigid

$$k = \infty$$

Space

$$k = 0$$

$\therefore \sigma = 0$] no stress generate



$$\sigma = \alpha T E$$

$$\frac{d^2y}{dx^2} = \frac{M}{EI} = \frac{1}{R} = \text{curvature}$$

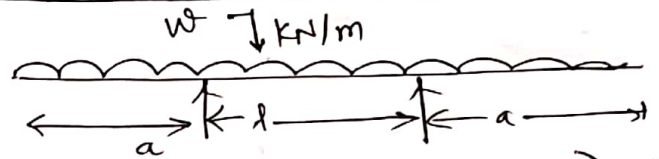
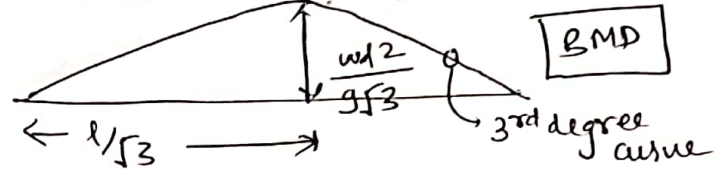
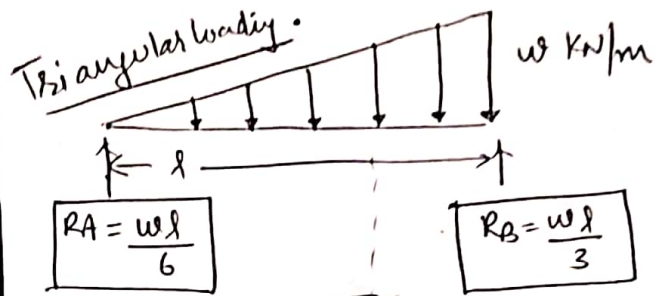
differentiate

radius of curvature

$$EI \frac{d^3y}{dx^3} = \frac{dM}{dx} = S \rightarrow \text{shear force}$$

differentiate

$$EI \frac{d^4y}{dx^4} = \frac{dS}{dx} = -w \rightarrow \text{KN/m Load intensity}$$



max. (+ve) BM = $\frac{w}{8} (l^2 - (2a)^2)$

max. (-ve) BM = $\frac{wa^2}{2}$

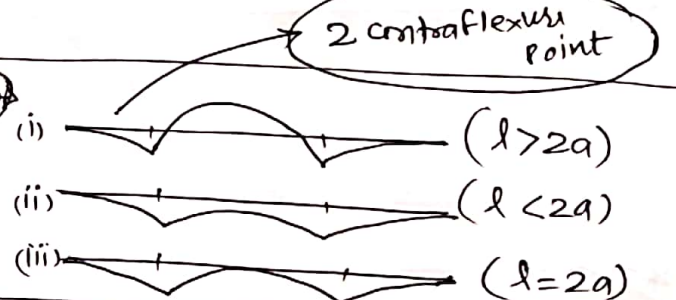
if both equal then

$a = \frac{l}{2\sqrt{2}}$

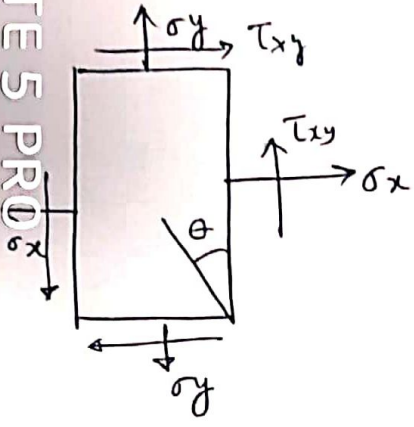
overhanging part = $\frac{a}{l+2a}$

(20.71%)

Slope of Bending diagram gives you $\rightarrow$ shear force.



Principal stress/strain :-



$$\sigma_{x'} = \left(\frac{\sigma_x + \sigma_y}{2} \right) + \left(\frac{\sigma_x - \sigma_y}{2} \right) \cos 2\theta + \tau_{xy} \sin 2\theta \quad (1)$$

$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y = \sigma_1 + \sigma_2$$

$$\sigma_{y'} \Rightarrow \text{put } \theta \rightarrow 90 + \theta \text{ in eqn (1)}$$

$$\sigma_1 / \sigma_2 = \left(\frac{\sigma_x + \sigma_y}{2} \right) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + (\tau_{xy})^2}$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

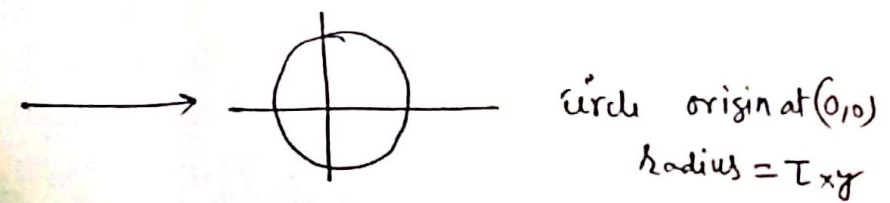
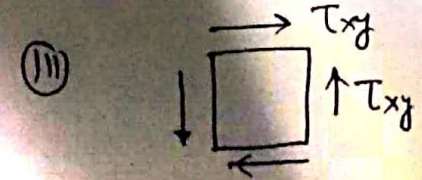
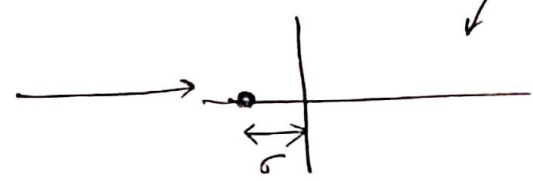
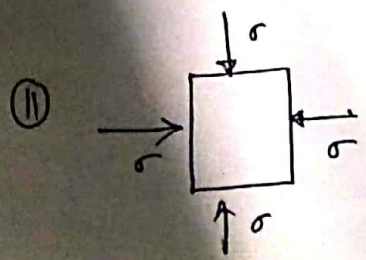
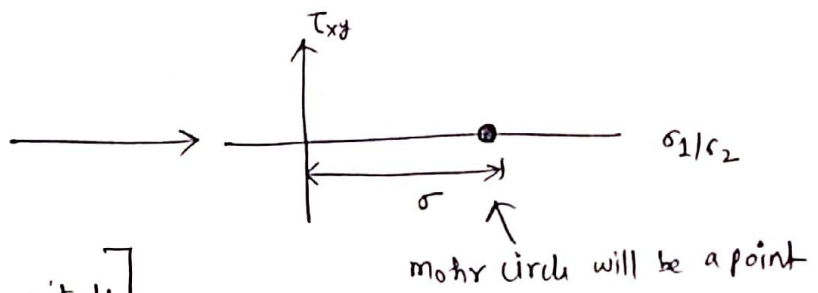
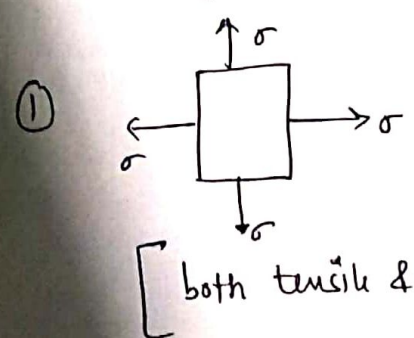
$\theta_p \rightarrow$ angle of principal plane

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Std. Mohr circle :-

$$\text{centru} = \left(\frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

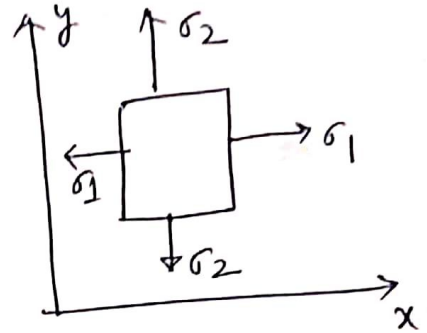
$$\text{radius} = \left| \frac{\sigma_1 - \sigma_2}{2} \right| = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + (\tau_{xy})^2}$$



Analysis of strain :- {when σ_1, σ_2 given}

major principal strain $\epsilon_1 = \frac{\sigma_1}{E} - \frac{\mu \sigma_2}{E}$

minor " " $\epsilon_2 = \frac{\sigma_2}{E} - \frac{\mu \sigma_1}{E}$



↓ solve then

$$\sigma_1 = \frac{E}{1-\mu^2} [\epsilon_1 + \mu \epsilon_2]$$

$$\sigma_2 = \frac{E}{1-\mu^2} [\epsilon_2 + \mu \epsilon_1]$$

• Mohr circle → Strain Analysis → put $T_{xy} \rightarrow \frac{\phi_{xy}}{2}$ rest will be same
 (to The point) $(\theta \rightarrow \text{from horizontal})$

total strain energy per unit volume = $\frac{1}{2} \sigma_1 \epsilon_1 + \frac{1}{2} \sigma_2 \epsilon_2 + \frac{1}{2} \sigma_3 \epsilon_3$

put $\epsilon_1 \& \epsilon_2$ in terms of $\sigma_1 \& \sigma_2$

3D
case

then

$$\frac{SE}{\text{volume}} \leftarrow U = \frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu (\sigma_1 \sigma_2 + \sigma_2 \sigma_3 + \sigma_3 \sigma_1) \right]$$

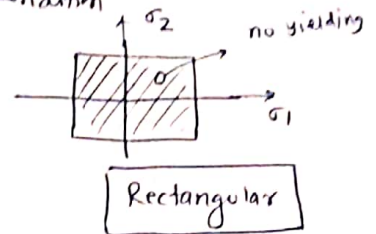
Theory of failure

for no failure condition

(i) maximum principal stress theory (Rankine Theory)

note:- This theory → Best for Brittle material
∵ Brittle material fails under tension (Brittle failure)

$$\sigma_{\max} \text{ or } \sigma_1 \leq (\sigma_y / FOS)$$



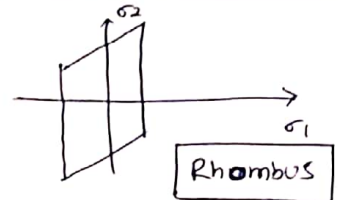
(ii) maximum principal strain theory (St. Venant's Theory)

note:- This theory overestimate the elastic strength of ductile material

Brittle + ductile

$$\epsilon_1 \leq \frac{(\sigma_y / FOS)}{E}$$

$$\left\{ \text{where } \epsilon_1 = \frac{\sigma_1}{E} - \nu \frac{\sigma_2}{E} - \nu \frac{\sigma_3}{E} \right\}$$



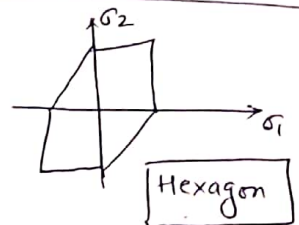
(iii) maximum shear stress theory (Guest & Tresca's theory)

note:-
• most conservative theory
• not for hydrostatic loading because $\sigma_1 = \sigma_2 = \sigma_3$
• ductile + shaft

$$\tau_{\max} \leq \frac{(\sigma_y / FOS)}{2}$$

$$\tau_{\max} = \left\{ \left| \frac{\sigma_1 - \sigma_2}{2} \right| \text{ or } \left| \frac{\sigma_1}{2} \right| \text{ or } \left| \frac{\sigma_2}{2} \right| \right\}$$

basically absolute max. shear stress



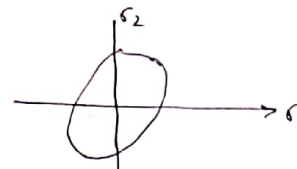
(iv) maximum strain energy theory (Haigh's theory)

note:- ductile + pressure vessel Best

$$SE / \text{volume} \leq \frac{(\sigma_y / FOS)^2}{2E}$$

$$\frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \right] \leq \frac{(\sigma_y / FOS)^2}{2E}$$

ellipse

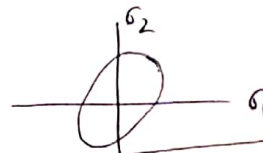


(v) max. shear strain energy theory (von-mises theory) / distortion theory

note:- ductile + pure shear

$$\frac{1}{12G} \left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \right] = \frac{(\sigma_y / FOS)^2}{6G}$$

ellipse



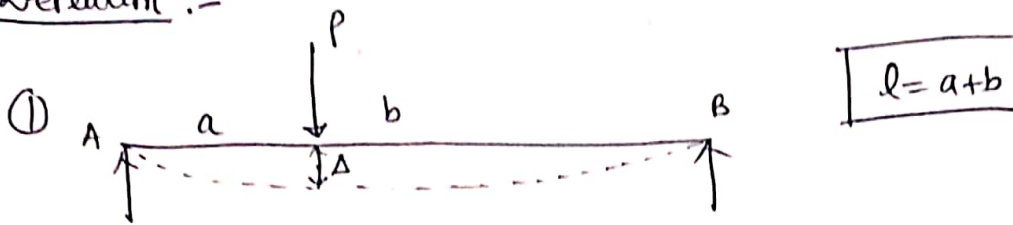
* Shear strain energy = $\frac{1}{2} \sigma_{avg} \epsilon_v$

→ Volumetric stress $\sigma_{avg} = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$

→ Volumetric strain $\epsilon_v = \frac{\sigma_1 + \sigma_2 + \sigma_3}{E} (1 - 2\nu)$

note:- when $\sigma_1 \gg \sigma_2, \sigma_3$

Deflection :-



$$l = a + b$$

$$\theta_A = \frac{Pab(l+b)}{6EI l}$$

$$\theta_B = \frac{Pab(l+a)}{6EI l}$$

$$\Delta_{\text{under load}} = \frac{Pa^2b^2}{3EI l}$$

$$\Delta_{\text{max}} = \frac{Pb(l^2 - b^2)^{3/2}}{9\sqrt{3}EI l}$$

$$a) \sqrt{\frac{l^2 - b^2}{3}} \left\{ \begin{array}{l} \Delta_{\text{max}} \rightarrow \text{between of} \\ \text{mid section of beam} \\ \text{and load} \end{array} \right\}$$

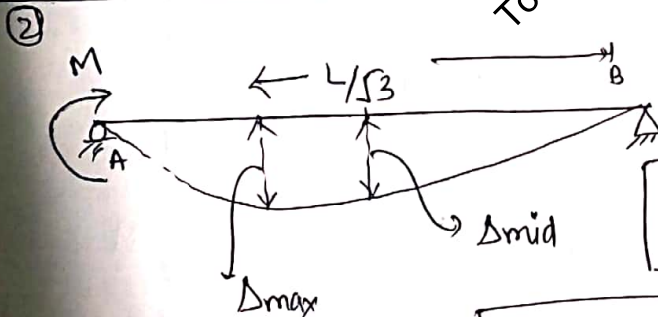


put $a = b = l/2$

$$\theta_A = \frac{Pl^2}{16EI} = \theta_B$$

$$\Delta = \frac{Pl^3}{48EI}$$

To The Point By DnyanPal



$$\theta_B = \frac{ML}{6EI}$$

$$\theta_A = \frac{ML}{3EI}$$

$$\Delta_{\text{mid}} = \frac{ML^2}{16EI}$$

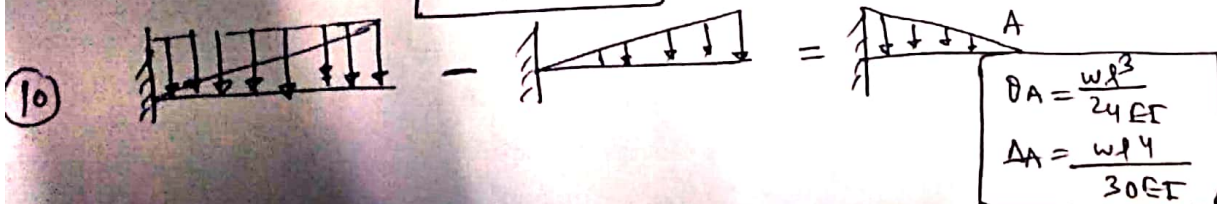
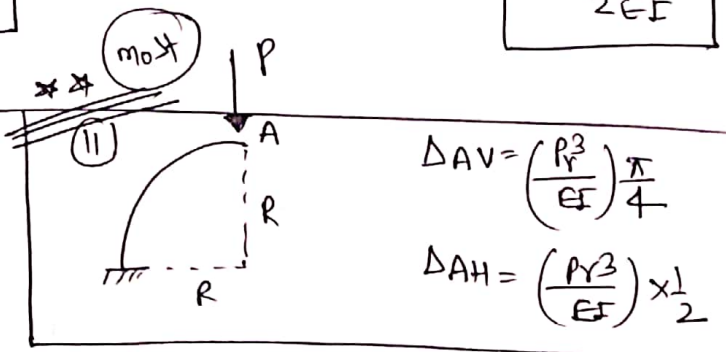
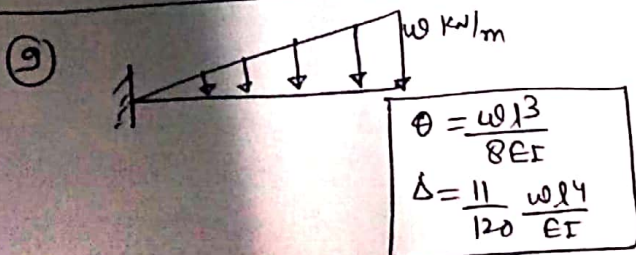
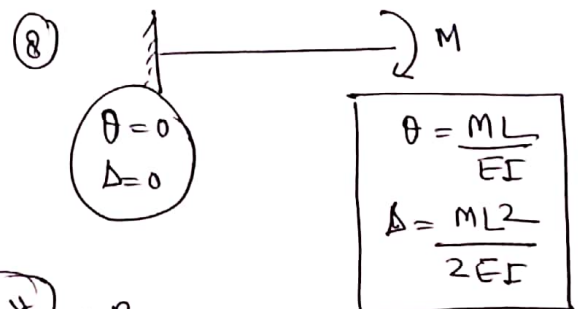
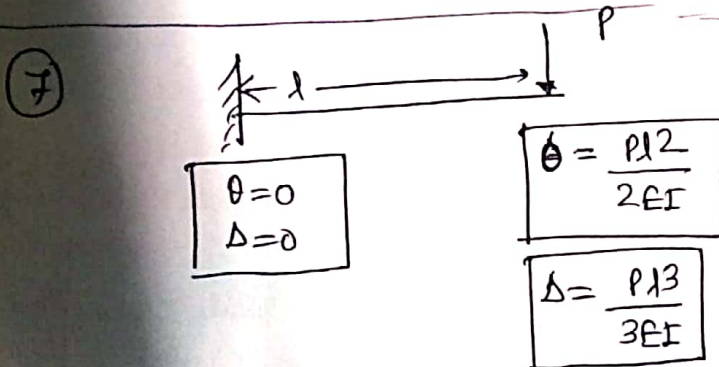
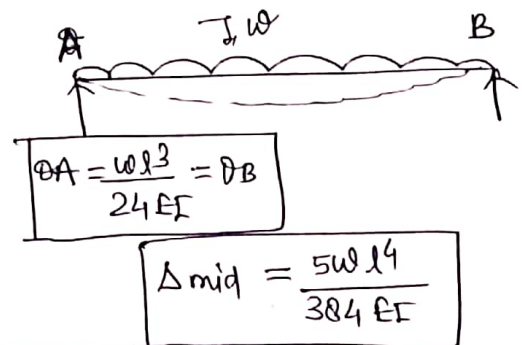
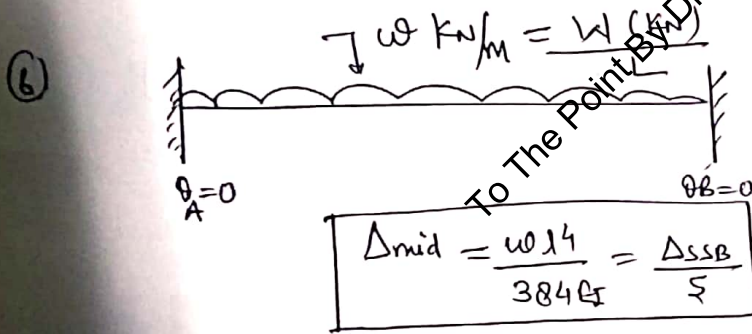
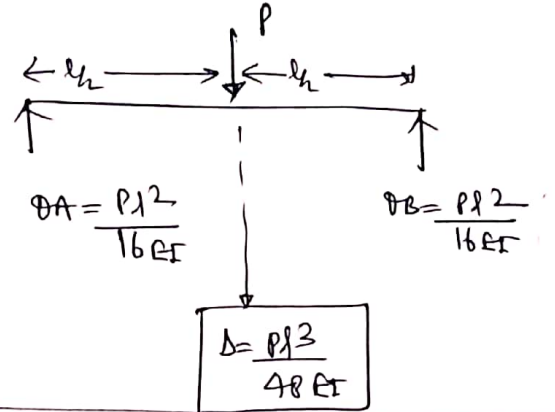
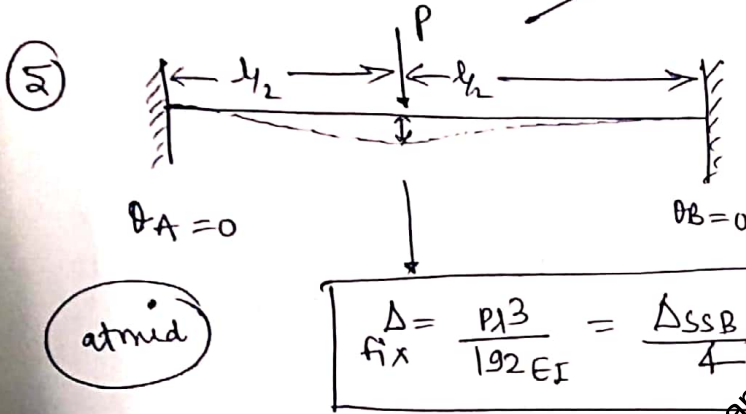
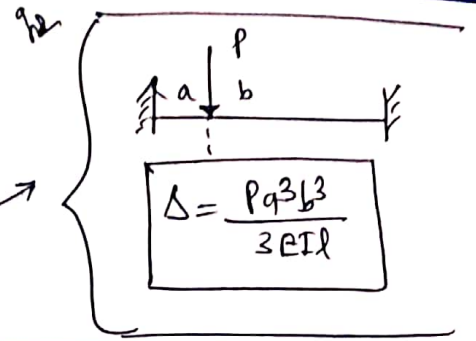
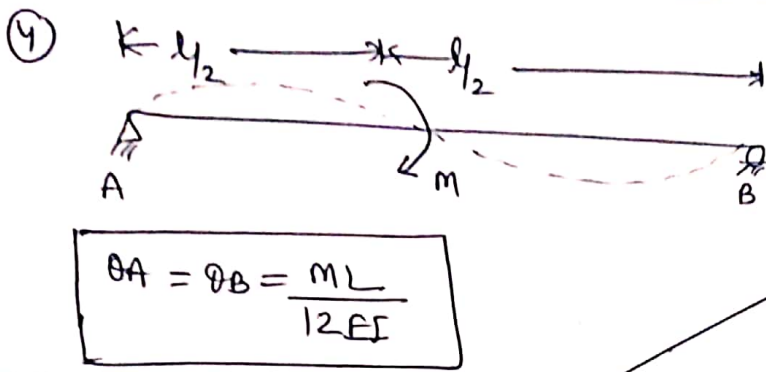
$$\Delta_{\text{max}} = \frac{ML^2}{9\sqrt{3}EI}$$

a) $L/\sqrt{3}$ from opposite end.

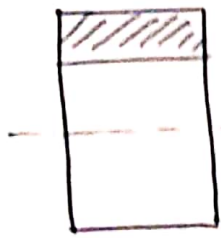


$$\theta_A = \theta_B = \frac{ML}{2EI}$$

$$\Delta_{\text{mid}} = \Delta_{\text{max}} = \frac{ML^2}{8EI}$$



Shear stress :-



SF (shear force) taken by this section

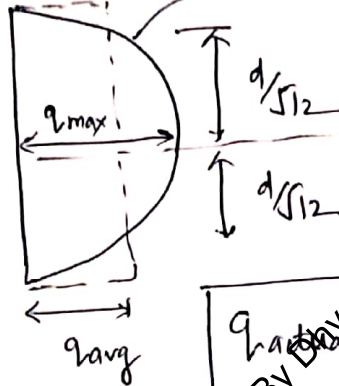
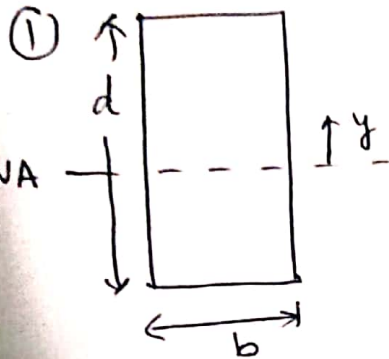
$$= \frac{\tau_{max}}{\gamma_{max}} (A\bar{y})$$

BM (Bending moment)

$$= \frac{\tau_{max}}{\gamma_{max}} (I_{area} \text{ about N.A.})$$

Std. result

$$Q = \frac{SA\bar{y}}{Ib}$$



$$q = \frac{6S}{bd^3} \left[\frac{d^2}{4} - y^2 \right]$$

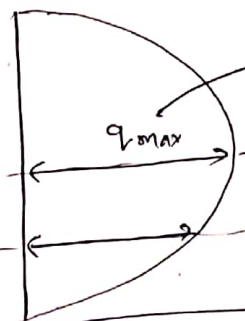
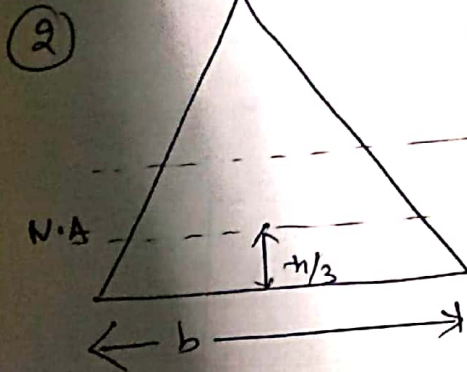
y → from N.A

$$q_{max} = q_{avg} \text{ @ } y = \frac{d}{\sqrt{12}}$$

at NA

$$q_{max} = \frac{3}{2} q_{avg}$$

$$q_{avg} = \frac{S}{bd}$$

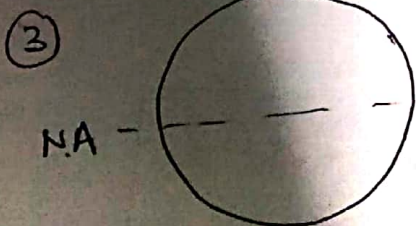


at mid height

$$q_{max} = \frac{3}{2} q_{avg}$$

$$q_{NA} = \frac{8}{3} \frac{S}{bh}$$

$$q_{avg} = \frac{S}{(\frac{1}{2}bh)}$$



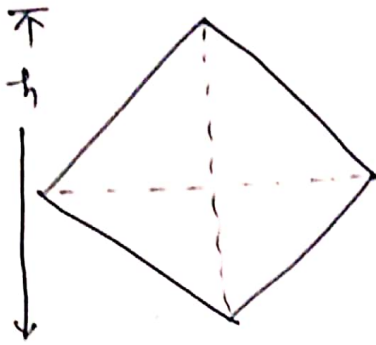
$$q_{max} = \frac{4}{3} q_{avg}$$

$$q_{avg} = \frac{S}{\pi R^2}$$

$$q = \frac{4}{3} \frac{S}{\pi R^4} (R^2 - y^2)$$

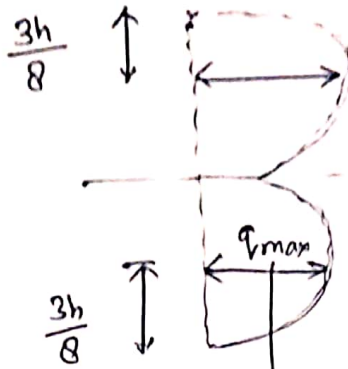
y → from N.A

④



[diagonal section]

[1 diagonal → vertical]



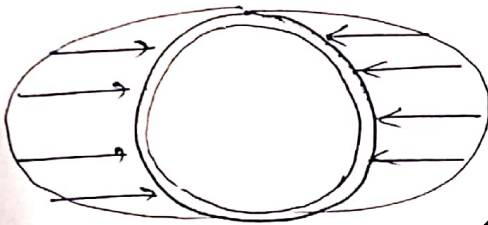
$$q_{max} = \frac{9}{8} q_{avg}$$

$$q_{avg} = \frac{S}{Area}$$

(from $\frac{3h}{8}$ dist from top and bottom)

⑤

Thin ~~thin~~ circular Tube :-



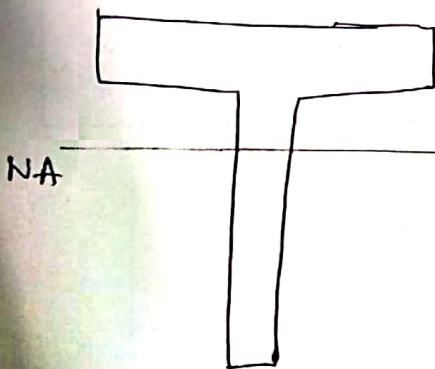
To The Point By DhyanPal

$$q_{max} = 2 q_{avg}$$

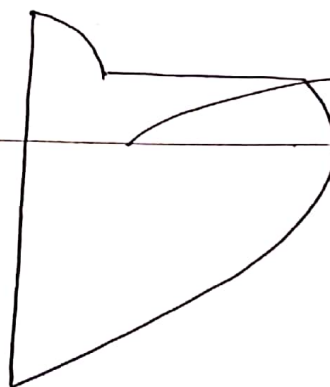
$$A = (2\pi R_m) t_m$$

$$I = \pi R_m^3 t_m$$

⑥

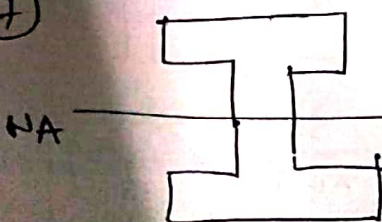


NA

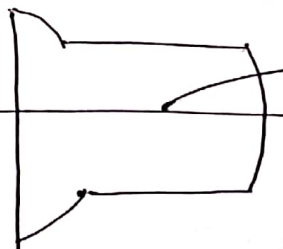


q_{max} at N.A

⑦



NA



q_{max} at NA

Torque :-

eqn of Torsion

$$I_P = I_{xx} + I_{yy}$$

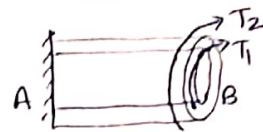
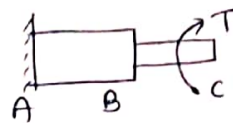
$$\frac{\tau}{r} = \frac{T}{I_P} = \frac{G\theta}{L}$$

valid for circular section.

Compound shaft

Series connection

Parallel connection



$$T_{AB} = T_{BC} = T$$

(torque same)

$$\theta_{AC} = \theta_{AB} + \theta_{BC}$$

$$\theta_{AB1} = \theta_{AB2}$$

angular deflection same

$$T = T_1 + T_2$$

$$\frac{GI_P}{L} \rightarrow \text{Torsional rigidity (GI)}$$

$$\frac{GI_P}{L} \rightarrow \text{Torsional stiffness}$$

ET $\rightarrow$ Flexural rigidity

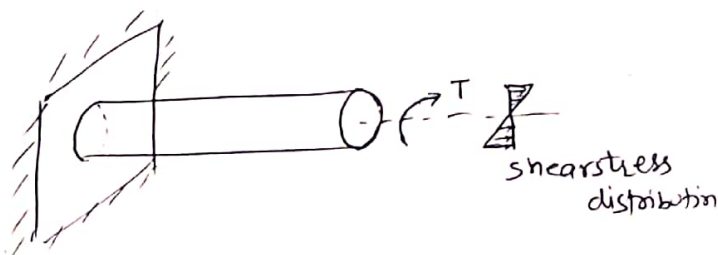
$\frac{EI}{L} \rightarrow$ Flexural stiffness

AE $\rightarrow$ axial rigidity

$\frac{AE}{L} \rightarrow$ axial stiffness.

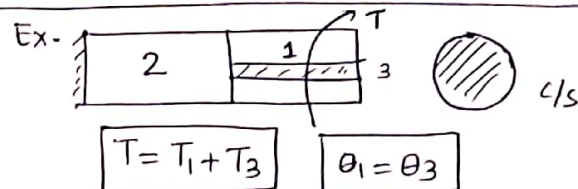
Assumption :- (1) circular section remains circular after twisting

- (ii) stress strain variation linear
- (iii) Axis of shaft loaded by twisting couple is perpendicular to plane of couple.



- (iv) twisting along shaft $\rightarrow$ uniform
- (v) plane section remain plain, does not warp.
- (vi) radial lines remains radial during torsion
- (vii) stress induced within elastic limit.

note: This topic $\rightarrow$ conventional / mains **



$$T = T_1 + T_3$$

$$\theta_1 = \theta_3$$

$$I_{P \text{ solid}} = \frac{\pi}{32} D^4$$

$$I_{xx} = \frac{\pi}{64} D^4$$

note: **

$$\text{tube } I_P = 2\pi R^3 t$$

$$I_{P \text{ hollow}} = \frac{\pi}{32} (D^4 - d^4)$$



$$P = \frac{2\pi NT}{60} \rightarrow \text{Nm}$$

(Watt)

Strain energy due to Torsion

$$\Rightarrow \frac{1}{2} T \theta$$

$$\Rightarrow \frac{1}{2} T \left[\frac{TL}{GI_P} \right]$$

$$\Rightarrow \frac{T^2 L}{2GI_P}$$

Same style like SE due to bending moment

$$= \frac{m^2 L}{2EI}$$

$$\text{Shear Resilience} = \frac{SE}{\text{Volume}} = \frac{1}{2} \tau \phi$$

Solid shaft

$$\frac{SE}{\text{Volume}} = \frac{\tau_{\max}^2}{4G\eta}$$

hollow shaft

$$\frac{SE}{\text{Volume}} = \frac{\tau_{\max}^2}{4G\eta} \left[\frac{R_o^2 + R_i^2}{R_o^2} \right]$$

$$\tau_{\max} = \frac{16T}{\pi D_s^3} \rightarrow \text{Solid shaft} \left\{ \frac{\tau}{r} = \frac{T}{I_P} \right\}$$

$$\tau_{\max} = \frac{16T}{\pi D^3 \left[1 - \frac{1}{\eta^4} \right]} \rightarrow \text{hollow shaft}$$

$$\eta = \frac{\text{outdia}}{\text{inner}} = \frac{D_o}{D_i}$$

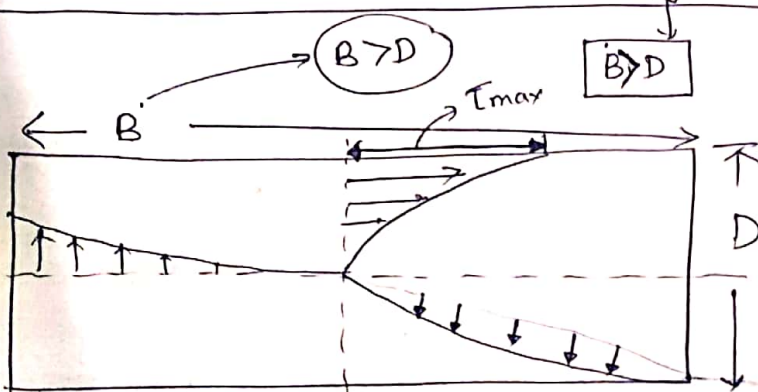
$$\frac{T_H}{T_S} = \frac{\eta + \frac{1}{\eta}}{\sqrt{\eta^2 - 1}}$$

torque comparison
b/w solid & hollow shaft

$$\frac{W_H}{W_S} = \frac{(\eta^2 - 1)\eta^{2/3}}{(\eta^4 - 1)^{2/3}}$$

weight comparison
b/w hollow solid shaft

Shear stress distribution for Rectangular Bar :-



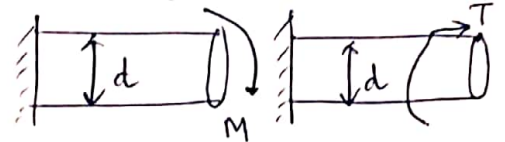
conclusion :- (i) Corners $\tau = 0$

(ii) $\tau_{\max}$ at boundary surface but overall max. $\rightarrow$ middle of longer side (B)

$$\text{Shear flow } q = \frac{T}{2\pi A m}$$

$$\text{Shear stress} \times \text{thickness} = q$$

(Solid shaft)



$$\tau = \frac{32M}{\pi d^3}$$

$$\tau_{\max} = \frac{16T}{\pi d^3}$$

$$\frac{\tau}{\tau_{\max}} = \frac{2M}{T}$$

combined effect of bending & torsion

$$\frac{\sigma_1}{\sigma_2} = \frac{\sigma_1}{\sigma_2} = \frac{16}{\pi D^3} [M \pm \sqrt{M^2 + T^2}]$$

Principal stresses $\{\sigma_1, \sigma_2\}$

$$\tau_{\max} = \frac{16}{\pi D^3} [\sqrt{M^2 + T^2}]$$

max. shear stress

$$\text{equivalent } Bm = \frac{1}{2} [M + \sqrt{M^2 + T^2}]$$

$$\frac{32M'}{\pi d^3} = \frac{16}{\pi d^3} (m + \sqrt{m^2 + t^2})$$

Same

$$\text{equivalent torque} = \sqrt{M^2 + T^2}$$

if M, T given then Shaft strength will be checked by this formula.

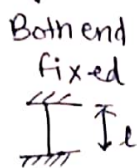
column :-

Euler Theory valid only
for long column

in this theory direct axial compressive stress neglected hence not used for short column.

$$P_{cr} = \frac{\pi^2 EI_{min}}{(l_{eff})^2} = P_e$$

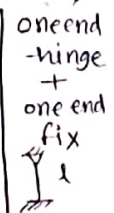
l_{eff}



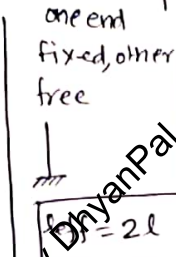
$$l_{eff} = \frac{l}{2}$$



$$l_{eff} = l$$



$$l_{eff} = \frac{l}{\sqrt{2}}$$



$$l_{eff} = 2l$$

Slenderness Ratio $\lambda = \frac{l_{eff}}{r_{min}}$ To The Point By DhyanPal min radius of gyration

$$r_{min} = \sqrt{\frac{I_{min}}{A}}$$

$$\therefore P_{cr} = \frac{\pi^2 EI_{min}}{\lambda^2 A_{min}^2} = \frac{\pi^2 EI_{min}}{\lambda^2 \times (I_{min}/A)}$$

$$\frac{P_{cr}}{A} = \sigma_{cr} = \text{euler critical stress or Buckling strength} = \frac{\pi^2 E}{\lambda^2}$$

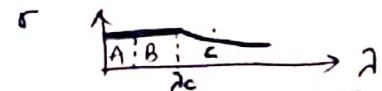
check :- validity of Euler's Theory

$$\sigma_{cr} \leq \sigma_y$$

$$\frac{\pi^2 E}{\lambda^2} \leq \sigma_y$$

$$\therefore \lambda \geq \sqrt{\frac{\pi^2 E}{\sigma_y}}$$

$\lambda_c = 90$ for mild steel



A $\rightarrow$ safe short column
B $\rightarrow$ safe long column
C $\rightarrow$ Intermediate safe column.

Rankine formula $\therefore$ valid $\rightarrow$ for short & long column
buckling & crushing considered

$$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_e}$$

crushing load

euler load

$$P_c = f_c A$$

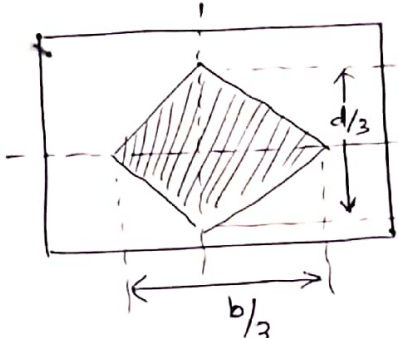
$$\therefore P_R = \frac{f_c A}{1 + \alpha \lambda^2}$$

$$\alpha = \text{Rankine constant} = \frac{f_c}{\pi^2 E}$$

Shape of kern in eccentric loading

① Rectangular & I section

Rhombus



for not tension $\left. \begin{array}{l} e < b/6 \\ e < d/6 \end{array} \right\} \therefore \text{called } \frac{1}{3} \text{rd Rule}$

② Square

Square

③ Circular

Circular



$e < d/8$
 $\therefore \frac{1}{4} \text{th rule}$

④ Hollow circular section

"

$$e < \frac{D^2 + d^2}{8D}$$

eccentricity

for design

(multiply by 2)

Spring

	closed coil	open coil
direct shear	✓	✓
Bending Stress	Neglected	✓
shear stress	✓	neglected

② strain energy stored in spring

$$U = \frac{1}{2} T \theta = \frac{1}{2} T \left(\frac{T L}{G I_p} \right)$$

यहाँ $T = PR$ $L = 2\pi R n$ $I_p = \frac{\pi d^4}{32}$

$$\therefore U = \frac{32 P^2 R^3 n}{G d^4}$$

③ axial deflection

$$\Delta = \frac{\partial U}{\partial P} = \frac{64 P R^3 n}{G d^4}$$

④

$$\text{stiffness } k = \frac{P}{\Delta} = \frac{G d^4}{64 R^3 n}$$

$$\text{flexibility} = \frac{1}{k}$$

⑤

Proof load :- max. load carrying capacity of spring without getting permanently distorted.

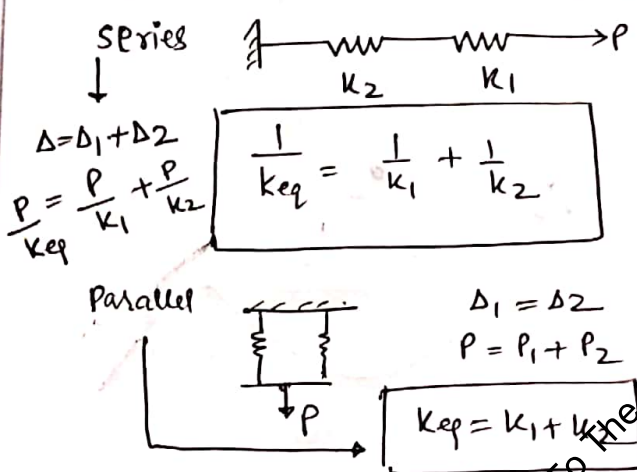
Let $f_s \rightarrow$ allowable shear stress of material of spring

$$\therefore f_s = \frac{16 (PR)}{\pi d^3}$$

$$\therefore P_{\max} = \text{Proof load} = \frac{\pi d^3 f_s}{16 R}$$

Spring

$\rightarrow$ Torque like capacitor



flexibility $= \frac{1}{k} = \Delta$ $\Delta = P/k$ $k \rightarrow \frac{N}{mm}$

closed coil :- (Bending stress $\rightarrow$ neglected)

①

$$T_{\max} = \text{innerside} = \frac{P}{\frac{\pi}{4} d^2} + \frac{16 (PR)}{\pi d^3}$$

direct stress shear stress due to torque

$$T_{\max} = \text{Inner} = \frac{16 PR}{\pi d^3} \left[1 + \frac{d}{4R} \right]$$

$$T_{\min} = \text{outer} = \frac{16 PR}{\pi d^3} \left[1 - \frac{d}{4R} \right]$$

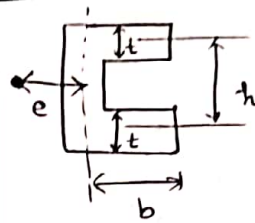
$\therefore \frac{d}{4R} \ll \ll \ll 1$ hence direct stress we can also neglect.

leaf spring :-

$$\text{max stress } f_{\max} = \frac{3 W l}{2 b n t^2}$$

Shear centre (or) (centre of flexure) $\odot S$ } $\rightarrow$ no torsion

Formula

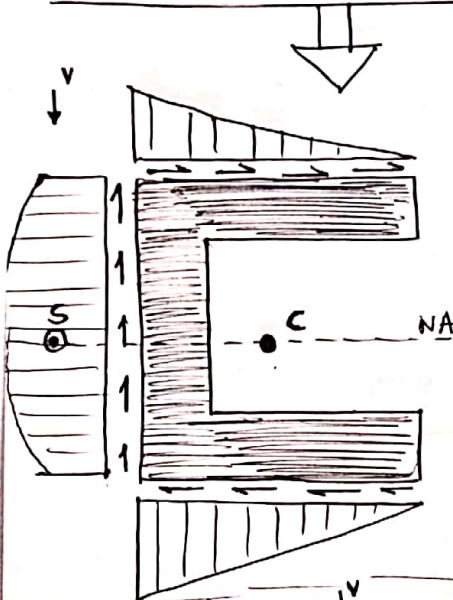


$$e = \frac{b^2 h^2 t}{4 I_{NA}}$$

Shear centre distance

Shear flow in common section for vertical load $\downarrow$ through section

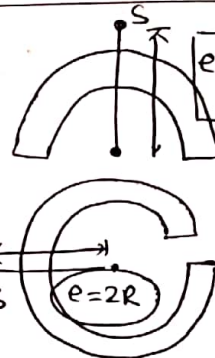
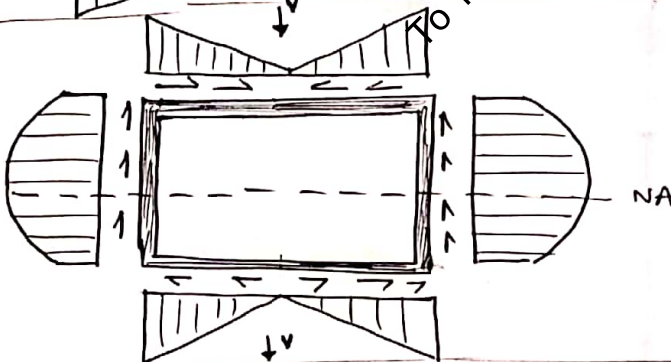
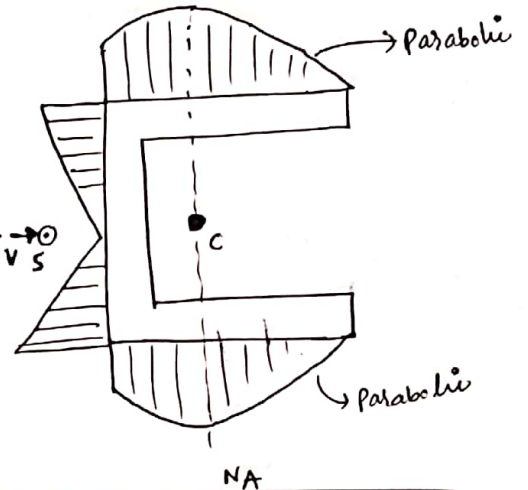
inf figure



Shear centre $\odot S$
Centroid $\bullet C$

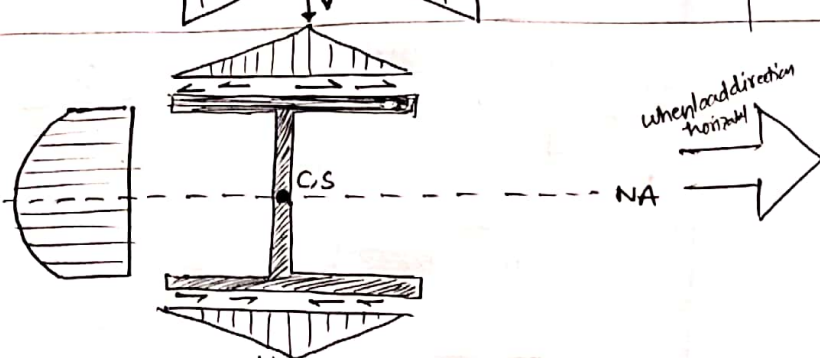
Shear flow tends to max. at NA and tends to zero at free edge.

when load direction horizontal



$$e = \frac{4R}{\pi}$$

Semi circular section



when load direction horizontal

